

YINING SHE

Pittsburgh, PA yiningsh@cs.cmu.edu sheyining.github.io

EDUCATION

Carnegie Mellon University

Ph.D. in Software Engineering, Advised by Eunsuk Kang
Software and Societal Systems Department, School of Computer Science
Research focus: Software Engineering for AI; Safety, Robustness & Fairness of AI System

Pittsburgh, PA

Aug.2022-May 2027(expected)

ShanghaiTech University

Bachelor of Engineering in Computer Science and Technology
GPA: **3.83/4.0**
Selected Honors: *Outstanding Graduate, Merit Student, Outstanding Teaching Assistant*

Shanghai, China

Sep.2018-Jun.2022

PROFESSIONAL SKILLS

Programming Languages

Python, C/C++, C#, R, MATLAB

Tools and Frameworks

PyTorch, scikit-learns, Tensorflow, OpenCV, Alloy, UPPAAL, Git, RISC-V

DOCTORAL RESEARCH

Safeguarding LLM Agent Action Through Provenance Identification

In-process

- Proposing a LLM agent runtime safeguard to prevent unintended consequences such as data leakage or financial harm caused by underspecified user intent and LLM hallucination
- Developing a provenance identification method to trace the source of proposed action and action parameters in the user query, background context, tool specification, and interaction history
- Incorporating multi-level provenance requirements into each tool, enabling strict control over high-impact actions (e.g., financial transactions or data-sharing requests) and flexible execution of others (e.g., non-sensitive information retrieval)
- Planning to evaluate the proposed safeguard in a customized agent safety benchmark, measuring reductions in unsafe behaviors compared to baselines and false positive interventions over safe actions

FAIRSENSE: Long-Term Fairness Analysis of ML-enabled Systems

In ICSE 2025

- Proposed a simulation-based framework FAIRSENSE to detect and analyze the long-term unfairness in ML-enabled systems
- Modeled feedback loop between an ML-enabled system and its deployment environment to evaluate fairness over time via Monte Carlo simulation.
- Performed sensitivity analysis on simulation traces to understand the impact of design options and environmental factors on the long-term fairness of the system
- Applied sampling heuristic to efficiently explore exponentially large configuration space without affecting sensitivity analysis results
- Evaluated FAIRSENSE on three real-world benchmarks: loan approval, opioid risk scoring, and predictive policing, and results show it can effectively detect long-term fairness issues and identify system configuration variables with the greatest impact for developers

INDUSTRY EXPERIENCE

Oracle Cloud Infrastructure

Redwood City, CA

GenAI Research Intern, Advised by Dan Roth (Paper in submission)

May 2025-Aug.2025

- Conducted a systematic evaluation study of 5 LLM-based guardrail models (including Llama Guards and GPTs) when facing RAG-style contexts
- Curated a RAG-context input and output guardrail evaluation benchmark using 6.7k harmful prompts collected from 7 benchmarks and 8 generation LLMs
- Identified input and output guardrail would be disturbed by RAG-style context in 11% and 8% cases respectively on average
- Analyzed how retrieved content, user query, and LLM response affects guardrail's robustness separately

- Explored how effective dedicated prompting and deep reasoning model can mitigate the issue

Intel Lab

Hillsboro, OR

AI Software Research Intern, Parallel Computing Lab

May 2024-Aug.2024

- Proposed an ML-based tensor kernel loop optimization method to predict the optimal loop configuration for a given machine
- Collected General Matrix Multiply (GEMM) kernel performance data across 1,040 loop tiling and ordering configurations on 15+ machines with diverse architectures and platforms
- Trained a multi-layer perceptron (MLP) model to predict kernel performance based on loop configuration and machine characteristics
- Integrated the ML model into a progressive ranked exploration pipeline that alternates between exploring new configurations and exploiting the current optimal solution
- Evaluated time efficiency of progressive ranked exploration compared to the baseline method, and results show it identifies the global optimal GEMM loop configuration for all 15+ machines by searching $\leq 3\%$ of the configuration space

PUBLICATIONS

- **Yining She**, Sumon Biswas, Christian Kästner, Eunsuk Kang. “**FairSense: Long-Term Fairness Analysis of ML-Enabled Systems**”, In the 47th IEEE/ACM International Conference on Software Engineering (ICSE), 2025.
- Sumon Biswas, **Yining She**, Eunsuk Kang. “**Towards Safe ML-based Systems in Presence of Feedback Loops**”, In *SE4SafeML, Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2023.
- Mohammad Piran, **Yining She**, Renzhi Tang, Zhihao Jiang, Yash Vardhan Pant. “**Stable Interaction of Autonomous Vehicle Platoons with Human-Driven Vehicles**”, In *American Control Conference (ACC)*, 2022.

TEACHING EXPERIENCE

TA for graduate course “Formal Methods”	<i>Carnegie Mellon University, Fall 2023</i>
Lead TA for undergraduate course “Algorithms and Data Structure”	<i>ShanghaiTech University, Fall 2021</i>
Lead TA for undergraduate course “Software Engineering”	<i>ShanghaiTech University, Spring 2021</i>
TA for undergraduate course “Algorithms and Data Structure”	<i>ShanghaiTech University, Fall 2020</i>